Feature Breakdown · Jira Tickets · Sprint Plan · Integration Map

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1. Feature Understanding & Analysis

1.1 Overview

The Support Dashboard enables the support team to operate a full ticketing system, track SLA compliance, collect learner satisfaction scores, and reduce resolution time through macros and templates. It integrates with every other dashboard — tickets can originate from Learner, Instructor, and Finance contexts — and feeds compliance evidence into the Admin Dashboard.

1.2 Five Core Feature Groups

F1 — Ticket Management Board	Full ticket lifecycle (New → In Progress → Resolved/Closed). Tickets carry priority (High / Medium / Low), category tags (Billing / Login / Certificate / Content / Technical), an assigned support agent, a linked course_id, and a threaded comments system. Tickets can be created by Learners, triggered by Instructors from the intervention queue, or raised internally by support staff.
F2 — SLA Engine	Configurable SLA policies per priority level. The engine tracks First Response Time (FRT) and Time To Resolution (TTR), flags breaches in real time, calculates backlog age, and sends breach-alert notifications to the assigned agent and their supervisor.
F3 — CSAT & NPS Surveys	On ticket close, a CSAT survey (1–5 stars) is sent automatically via Resend (email). NPS surveys (0–10) are sent on a configurable cadence per course. Results are segmented per course and per channel (email / SMS / WhatsApp). Follow-up automations trigger for detractor scores. Unifonic/Infobip handles SMS and WhatsApp delivery.
F4 — Macros & Templates	Support agents can apply pre-built reply macros to tickets with one click. Macros support variable substitution (learner name, course name, ticket ID). Escalation paths define which role receives a ticket after N hours without response. Knowledge base articles are linked within macros for self-service deflection.
F5 — Reports & Exports	Daily and weekly ticket volume reports with breakdown by tag, priority, and agent. Top recurring issues with frequency heatmap. SLA compliance rate. CSAT/NPS trend charts. All reports exportable as CSV and PDF with timestamp and filter context for NELC audit evidence bundles.

1.3 Cross-Dashboard Touch Points

Dashboard	Integration
Learner Dashboard	Open ticket with priority; view ticket status, SLA timer, and agent reply thread.
Instructor Dashboard	Open ticket from At-Risk Learner intervention queue; ticket pre-tagged with learner + course co
Course Manager	View count of open tickets per course; flag high-volume issue courses.
Finance Dashboard	Billing-tagged tickets link to invoice/payment records; refund requests can auto-generate a ticke
Admin Dashboard	Support KPIs widget: FRT/TTR averages, CSAT/NPS scores, breach rate, issues heatmap by

2. Data Model & Database Schema

Table: support_tickets

```
ticket_id (PK, UUID) | user_id (FK→users) | course_id (FK→courses, nullable) | invoice_id (FK→invoices, nullable) | subject VARCHAR(255) | status ENUM(new, in_progress, on_hold, resolved, closed) | priority ENUM(high, medium, low) | category ENUM(billing, login, certificate, content, technical) | assigned_to (FK→users, nullable) | first_response_at TIMESTAMP | resolved_at TIMESTAMP | closed_at TIMESTAMP | sla_breached_frt BOOL | sla_breached_ttr BOOL | created_at | updated_at
```

Table: ticket_comments

```
comment_id (PK) | ticket_id (FK) | author_id (FK→users) | body TEXT | is_internal BOOL (private note vs public reply) | created_at
```

Table: sla_policies

```
policy_id (PK) | priority ENUM(high, medium, low) | frt_minutes INT | ttr_minutes INT | escalate_after_minutes INT | escalate_to_role VARCHAR | active BOOL | created_at
```

Table: sla_events

```
event_id (PK) | ticket_id (FK) | event_type ENUM(frt_breach, ttr_breach, escalation_triggered, resolved_within_sla) | triggered_at TIMESTAMP | notified_user_ids JSON
```

Table: support_macros

```
macro_id (PK) | title VARCHAR | body_template TEXT (supports {{learner_name}}, {{course_name}}, {{ticket_id}} variables) | kb_article_urls JSON | escalation_path_id (FK, nullable) | created_by (FK→users) | updated_at
```

Table: escalation_paths

```
path_id (PK) | name VARCHAR | steps JSON (array: {after_minutes, assign_to_role}) | active BOOL
```

Table: csat_surveys

```
survey_id (PK) | ticket_id (FK) | user_id (FK) | course_id (FK, nullable) | score TINYINT (1–5) | comment TEXT | channel ENUM(email, sms, whatsapp) | sent_at TIMESTAMP | submitted_at TIMESTAMP
```

Table: nps_surveys

```
nps_id (PK) | user_id (FK) | course_id (FK, nullable) | score TINYINT (0–10) | comment TEXT | channel ENUM(email, sms, whatsapp) | sent_at | submitted_at
```

Table: kb_articles

```
article_id (PK) | title_ar VARCHAR | title_en VARCHAR | body_ar TEXT | body_en TEXT | category ENUM(billing, login, certificate, content, technical) | published BOOL | view_count INT | created_at | updated_at
```

3. API Endpoints Specification

Ticket Management

Method	Endpoint	Description
POST	`/api/v1/tickets`	Create ticket (any authenticated role)
GET	`/api/v1/tickets`	List tickets (filters: status, priority, tag, assigned_to, course_id, date)
GET	`/api/v1/tickets/:id`	Get ticket detail with comment thread
PATCH	`/api/v1/tickets/:id`	Update status / priority / assignment
POST	`/api/v1/tickets/:id/comments`	Add comment (public reply or internal note)
POST	`/api/v1/tickets/:id/resolve`	Resolve ticket → triggers CSAT send
POST	`/api/v1/tickets/:id/close`	Close ticket
POST	`/api/v1/tickets/:id/escalate`	Manual escalation to supervisor
POST	`/api/v1/tickets/:id/apply-macro`	Apply macro template to ticket reply

SLA Engine

Method	Endpoint	Description
GET	`/api/v1/sla/policies`	List SLA policies by priority
PUT	`/api/v1/sla/policies/:priority`	Update FRT/TTR targets (Admin only)
GET	`/api/v1/sla/kpis`	Aggregated FRT avg, TTR avg, breach rate
GET	`/api/v1/sla/breaches`	Active breaches list with ticket context
GET	`/api/v1/sla/backlog`	Open tickets by age bucket (0-1h, 1-4h, 4-24h, 24h+)

CSAT & NPS

Method	Endpoint	Description
POST	`/api/v1/surveys/csat`	Submit CSAT score (public, token-authenticated)
POST	`/api/v1/surveys/nps`	Submit NPS score (public, token-authenticated)
GET	`/api/v1/surveys/summary`	Aggregated CSAT/NPS by course, channel, date
POST	`/api/v1/surveys/send-nps`	Trigger NPS batch send (Admin/Support)

Macros & Templates

Method	Endpoint	Description
GET	`/api/v1/macros`	List macros
POST	`/api/v1/macros`	Create macro
PATCH	`/api/v1/macros/:id`	Update macro
DELETE	`/api/v1/macros/:id`	Delete macro
GET	`/api/v1/escalation-paths`	List escalation path configs

Knowledge Base

Method	Endpoint	Description
GET	`/api/v1/kb/articles`	List articles (public + filtered by category)
GET	`/api/v1/kb/articles/:id`	Get article detail (bilingual)
POST	`/api/v1/kb/articles`	Create KB article (Support/Admin)
PATCH	`/api/v1/kb/articles/:id`	Update article

Reports

Method	Endpoint	Description
GET	`/api/v1/reports/support/volume`	Daily/weekly ticket volume by tag/priority
GET	`/api/v1/reports/support/issues`	Top recurring issues with frequency
GET	`/api/v1/reports/support/sla-compliance`	SLA breach rate by period
GET	`/api/v1/reports/support/agent-performance`	Per-agent FRT/TTR/CSAT
GET	`/api/v1/reports/support/export`	CSV/PDF export with filters + timestamp

Cross-Dashboard Reads

Method	Endpoint	Description
GET	`/api/v1/tickets/my-tickets`	Learner: own ticket list + status + SLA timer
GET	`/api/v1/tickets/course/:courseId/count`	Course Manager: open ticket count per course
GET	`/api/v1/admin/support-kpis`	Admin widget: FRT/TTR/CSAT/NPS/breach summary

4. Cross-Dashboard Integration Map

Every integration below has a defined data flow direction, the triggering event, and the API endpoint involved. All cross-dashboard reads are RBAC-gated.

Source	Trigger	Direction	Target	Endpoint / Action
Learner Dashboard	Learner clicks 'Open Ticket'	→	Support Dashboard	POST /api/v1/tickets course_id auto-populated
Learner Dashboard	Learner views My Tickets	→	Support Dashboard	GET /api/v1/tickets/my-tickets Shows status + SLA timer
Instructor Dashboard	Instructor opens ticket from At-Risk queue	→	Support Dashboard	POST /api/v1/tickets Pre-tagged: learner_id + course_id + category=cont
Finance Dashboard	Refund request raised	→	Support Dashboard	POST /api/v1/tickets invoice_id linked, category=billing, priority=high
Support Dashboard	Ticket resolved	→	Notification System	Resend email CSAT survey + Unifonic SMS/WhatsApp
Support Dashboard	SLA breach detected	→	Support Dashboard (agent)	Resend email breach alert + in-app notification
Admin Dashboard	Loads support KPIs widget	←	Support Dashboard	GET /api/v1/admin/support-kpis FRT, TTR, CSAT, NPS, breach rate
Course Manager	Views course details	←	Support Dashboard	GET /api/v1/tickets/course/:id/count Open ticket count badge

5. Notification System (Resend + Unifonic/Infobip)

Resend is the primary email provider (already integrated). Unifonic or Infobip handles SMS and WhatsApp. A unified NotificationService abstraction routes each event to the correct channel based on user preference and event type.

Event	Channel	Provider	Template	Trigger
Ticket Created (learner confirmation)	Email	Resend	ticket_created_learner	POST /tickets
Ticket Assigned to Agent	Email + In-App	Resend	ticket_assigned_agent	PATCH /tickets/:id
New Public Reply on Ticket	Email	Resend	ticket_reply_notification	POST /tickets/:id/comments (is_internal=fa
SLA FRT Breach Alert	Email + In-App	Resend	sla_frt_breach	SLA Engine (cron every 5 min)
SLA TTR Breach Alert	Email + SMS	Resend + Unifonic	sla_ttr_breach	SLA Engine (cron every 5 min)
Ticket Resolved	Email	Resend	ticket_resolved	POST /tickets/:id/resolve
CSAT Survey (post-resolution)	Email + WhatsApp	Resend + Unifonic	csat_survey	15 min after resolve (async job)
NPS Survey (periodic)	Email + SMS	Resend + Unifonic	nps_survey	Configurable cadence (cron)
CSAT Detractor Follow-up	Email	Resend	csat_followup	Score <= 2: auto-trigger
Escalation Triggered	Email + In-App	Resend	ticket_escalation	Escalation path timer fires

5.1 Resend Templates Required

- ticket_created_learner — Ticket #{id} received, expected response within {frt_target}
- ticket_assigned_agent — You have been assigned ticket #{id} (Priority: {priority})
- ticket_reply_notification — New reply on your ticket #{id}
- sla_frt_breach — BREACH ALERT: Ticket #{id} has exceeded FRT target ({frt_target} min)
- sla_ttr_breach — BREACH ALERT: Ticket #{id} has exceeded TTR target ({ttr_target} min)
- ticket_resolved — Your issue has been resolved. Please rate your experience.
- csat_survey — 1-click CSAT rating (embedded link tokens, 5 star options)
- nps_survey — How likely are you to recommend this course? (0–10 scale, embedded link)
- csat_followup — We're sorry your experience didn't meet expectations. Can we help further?
- ticket_escalation — Ticket #{id} has been escalated to {escalate_to_role}

6. Sprint Plan (5 Sprints)

Sprint	Theme	Story Points	Key Deliverables
Sprint 1	Foundation (Data Model + Ticket CRUD)	34 SP	DB migrations, Ticket CRUD API, Tickets Board UI (Kanban), role-based ac
Sprint 2	SLA Engine	26 SP	SLA policy table, FRT/TTR timers, breach detection cron, breach alert notifi
Sprint 3	Notification System	22 SP	10 Resend templates, Unifonic/Infobip SMS+WhatsApp adapter, CSAT surv
Sprint 4	CSAT/NPS + Macros & Knowledge Base	28 SP	CSAT/NPS results dashboard, Macros CRUD + apply, Escalation path conf
Sprint 5	Reports + Cross-Dashboard Integrations	30 SP	Volume/issues/SLA/agent reports, CSV+PDF exports, Admin KPI widget, Le

Total estimated effort: 140 Story Points across 5 sprints (2-week sprints = 10-week delivery window).

7. Jira Tickets — Sprint 1: Foundation

PHYSIO-SUP-01 — **Database migrations: support_tickets, ticket_comments, sla_policies, sla_events tables** — SP: 5 | Sprint 1

Assignee: Backend Lead	**Priority: High**	Tags: Support / Backend

Description:

- Create support_tickets table with all fields (status enum, priority enum, category enum, sla_breached flags)
- Create ticket_comments table with is_internal flag for private notes
- Create sla_policies table with frt_minutes, ttr_minutes, escalate_after_minutes per priority
- Create sla_events table for audit log of all SLA events
- Add foreign keys: user_id, course_id (nullable), invoice_id (nullable), assigned_to (nullable)
- Seed default SLA policies: High=FRT 30min/TTR 4h, Medium=FRT 2h/TTR 24h, Low=FRT 8h/TTR 72h

Acceptance Criteria:

1. All tables created with correct column types and constraints
2. Migrations reversible (down migration exists)
3. Default SLA seed data present after migration
4. All FK relationships enforced at DB level

PHYSIO-SUP-02 — **Database migrations: support_macros, escalation_paths, kb_articles, csat_surveys, nps_surveys tables** — SP: 3 | Sprint 1

Assignee: Backend Lead	**Priority: High**	Tags: Support / Backend

Description:

- Create support_macros table with body_template (variable substitution support), kb_article_urls JSON
- Create escalation_paths table with steps as JSON array
- Create kb_articles table with bilingual columns (title_ar, title_en, body_ar, body_en)
- Create csat_surveys and nps_surveys tables with channel enum

Acceptance Criteria:

1. All 5 tables created with correct schema
2. kb_articles bilingual columns properly nullable
3. survey tables have unique constraint on (ticket_id, user_id) for CSAT

PHYSIO-SUP-03	POST /api/v1/tickets — Create ticket endpoint (all authenticated roles)	SP: 5 | Sprint 1
Assignee: Backend Lead	Priority: High	Tags: Support / Backend / API

Description:

- Authenticated endpoint accessible to Student, Instructor, Support, Admin roles
- Required fields: subject, category, priority (defaults to medium if not provided)
- Optional fields: course_id, invoice_id (for billing tickets from Finance flow)
- Auto-set: user_id from JWT, status=new, created_at
- On creation: trigger ticket_created_learner Resend email notification (async)
- Return: full ticket object including generated ticket_id

Acceptance Criteria:

1. Unauthenticated requests return 401
2. Missing required fields return 422 with field-level errors
3. ticket_id is UUID and unique
4. Resend email dispatched within 30 seconds of creation
5. course_id FK validated if provided

PHYSIO-SUP-04	GET /api/v1/tickets — List tickets with filters (Support/Admin) and GET /api/v1/tickets/my-tickets (Learner)	SP: 5 | Sprint 1
Assignee: Backend Lead	Priority: High	Tags: Support / Backend / API

Description:

- Support role sees all tickets; Learner role sees only own tickets
- Filter params: status, priority, category, assigned_to, course_id, date_from, date_to, search (subject text)
- Pagination: page + per_page with total count
- GET /my-tickets: returns learner's own tickets with status, assigned agent name, SLA timer remaining
- Response includes: ticket_id, subject, status, priority, category, assigned_to, created_at, first_response_at, sla_breached flags

Acceptance Criteria:

1. Learner cannot see other users' tickets (403 if attempted)
2. Filter combinations return correct subsets
3. Pagination headers present (X-Total-Count, X-Page)
4. SLA remaining time calculated correctly in my-tickets response

PHYSIO-SUP-05	GET /api/v1/tickets/:id — Ticket detail with comment thread	SP: 3 | Sprint 1
Assignee: Backend Lead	Priority: High	Tags: Support / Backend / API

Description:

- Returns full ticket object with array of comments (oldest first)
- Internal notes (is_internal=true) visible only to Support/Admin roles, not to Learner/Instructor
- Includes assigned agent display name and avatar URL
- Includes SLA status: frt_status (met/breached/pending), ttr_status, time_remaining_minutes

Acceptance Criteria:

1. Learner accessing another user's ticket returns 403
2. Internal notes stripped from response when requester is Student role
3. SLA time_remaining_minutes is negative if breached

PHYSIO-SUP-06	PATCH /api/v1/tickets/:id — Update status/priority/assignment + POST /api/v1/tickets/:id/comments	SP: 5 | Sprint 1

Assignee: Backend Lead	Priority: High	Tags: Support / Backend / API

Description:

- PATCH: Support/Admin can update status, priority, assigned_to, category
- Status transition on first assignment: set first_response_at if not already set
- Status transition to resolved: set resolved_at, trigger CSAT send job (15 min delay)
- POST comments: Support/Admin can post public replies or internal notes (is_internal flag)
- POST comments by Learner: always public (is_internal=false forced)
- On new public reply: trigger ticket_reply_notification to the other party via Resend

Acceptance Criteria:

1. first_response_at set only once (on first Support reply, not on reassignment)
2. Learner cannot update ticket status or priority
3. Internal note does NOT trigger reply notification to learner
4. CSAT job queued correctly on resolve (check job queue table)

PHYSIO-SUP-07	Support Dashboard — Tickets Board UI (Kanban: New / In Progress / Resolved / Closed)	SP: 8 | Sprint 1

Assignee: Frontend Lead	Priority: High	Tags: Support / Frontend / UI

Description:

- Kanban board with 4 columns: New, In Progress, Resolved, Closed
- Ticket cards show: ID, subject (truncated), priority badge (colour-coded), category tag, assigned agent avatar, SLA time remaining (green/amber/red)
- Priority badges: High=red, Medium=amber, Low=green
- Drag-and-drop to move ticket between status columns (Support role only)
- Filter bar: priority, category, assigned agent, course, date range
- Search by ticket ID or subject keyword
- Bilingual support: Arabic RTL / English LTR toggle
- WCAG 2.1 AA: colour contrast, keyboard navigation, focus indicators

Acceptance Criteria:

1. Board renders correctly in both AR (RTL) and EN (LTR) layouts
2. Drag-and-drop updates status via PATCH API and shows optimistic UI
3. SLA timer colour changes: green > 50% time remaining, amber 10–50%, red < 10% or breached
4. All interactive elements reachable via keyboard
5. Colour contrast ratio >= 4.5:1 for all text on badge backgrounds

PHYSIO-SUP-08	Support Dashboard — Ticket Detail View (full thread + actions panel)	SP: 5 | Sprint 1
Assignee: Frontend Lead	Priority: High	Tags: Support / Frontend / UI

Description:

- Ticket header: ID, subject, status badge, priority, category, course (if linked), created_at
- SLA panel: FRT status, TTR status, time remaining countdown, breach indicator
- Comment thread: public replies and internal notes (notes shown with grey background to Support)
- Reply composer: rich-text input, toggle Public/Internal, Send button
- Actions panel: change status, reassign agent, change priority, apply macro, escalate
- Linked records: if invoice_id present, show invoice summary card

Acceptance Criteria:

1. Internal notes not visible to learners (tested with learner session)
2. Apply macro fills reply composer with template text (variables substituted)
3. Status change reflected immediately without page reload

PHYSIO-SUP-09	Learner Dashboard — Open Ticket widget + My Tickets list	SP: 5 | Sprint 1
Assignee: Full-Stack Dev	Priority: Medium	Tags: Support / Learner Dashboard / Frontend

Description:

- Add 'Support & Help' section to Learner Dashboard (as per SOW)
- Open Ticket form: subject, category (dropdown), description, optional course context (pre-populated if opened from course page)
- My Tickets list: ticket ID, subject, status badge, last updated, action button (View / Reply)
- Ticket status shown with SLA timer if open
- FAQs section: links to top KB articles by category

Acceptance Criteria:

1. Creating ticket from course page auto-populates course_id
2. Learner can only view own tickets
3. Form validates required fields before submission
4. Accessible: form labels linked to inputs, error messages announced via aria-live

8. Jira Tickets — Sprint 2: SLA Engine

PHYSIO-SUP-10	SLA Engine: FRT/TTR timer tracking service + cron job (every 5 minutes)	SP: 8 | Sprint 2
Assignee: Backend Lead	Priority: Critical	Tags: Support / Backend / SLA

Description:

- Cron job runs every 5 minutes to evaluate all open tickets against their SLA policy
- FRT breach: ticket has no first_response_at AND (now - created_at) > policy.frt_minutes
- TTR breach: ticket not resolved AND (now - created_at) > policy.ttr_minutes
- On breach detection: set sla_breached_frt=true / sla_breached_ttr=true, insert into sla_events, dispatch notification
- Breach notification dispatched only once per ticket per breach type (idempotent)
- Escalation check: if (now - created_at) > escalation_path.steps[n].after_minutes, reassign ticket

Acceptance Criteria:

1. Cron does not re-send breach notification if already sent (sla_events dedup check)
2. SLA policy correctly selected based on ticket.priority
3. Escalation chain executes steps in order
4. Performance: cron completes within 30 seconds for 10,000 open tickets

PHYSIO-SUP-11	GET /api/v1/sla/kpis + GET /api/v1/sla/breaches + GET /api/v1/sla/backlog	SP: 5 | Sprint 2
Assignee: Backend Lead	Priority: High	Tags: Support / Backend / API / SLA

Description:

- GET /sla/kpis: returns avg_frt_minutes, avg_ttr_minutes, breach_rate_percent, tickets_resolved_within_sla, period filter
- GET /sla/breaches: returns list of currently open tickets with active breach, includes ticket details and breach type
- GET /sla/backlog: returns open tickets bucketed by age: 0–1h, 1–4h, 4–24h, 24h+, with count per bucket
- All endpoints Support/Admin role only

Acceptance Criteria:

1. avg_frt calculation excludes tickets where first_response_at is null (unresponded)
2. Breaches list updates within 5 minutes of breach occurrence (cron latency)
3. Backlog buckets sum equals total open ticket count

PHYSIO-SUP-12	SLA KPIs widget on Support Dashboard UI (FRT, TTR, breach alerts, backlog age)	SP: 5 | Sprint 2
Assignee: Frontend Lead	Priority: High	Tags: Support / Frontend / SLA / UI

Description:

- KPI cards row: Avg FRT (vs target), Avg TTR (vs target), Active Breaches count, Backlog Age distribution
- Active Breaches count shown in red badge if > 0
- Backlog age shown as horizontal bar chart: 4 age buckets, colour-coded
- Breach alerts panel: scrollable list of breached tickets with ticket ID, subject, breach type, time overdue
- Auto-refresh every 60 seconds
- All KPIs filterable by date range and priority

Acceptance Criteria:

1. FRT/TTR cards show green if avg < 80% of target, amber 80–100%, red > target
2. Breach alerts panel updates without full page reload
3. Bilingual labels (AR/EN)

PHYSIO-SUP-13	SLA Policy Configuration UI (Admin) — PUT /api/v1/sla/policies/:priority	SP: 5 | Sprint 2
Assignee: Full-Stack Dev	Priority: Medium	Tags: Support / Admin / Backend / Frontend

Description:

- Admin Settings page: SLA Policies section
- Table showing current FRT/TTR/escalation targets per priority (High/Medium/Low)
- Edit mode: inline editing of frt_minutes, ttr_minutes, escalate_after_minutes
- Save triggers PUT endpoint which updates policy and reseeds any cached values
- Escalation path configuration: add/remove steps (after_minutes, assign_to_role)

Acceptance Criteria:

1. Only Admin role can access this settings page
2. Updated policies take effect on next cron run
3. Escalation path steps validated: each step must have after_minutes > previous step

PHYSIO-SUP-14	Breach alert notifications — Resend email + in-app notification on FRT and TTR breach	SP: 5 | Sprint 2
Assignee: Backend Lead	Priority: High	Tags: Support / Backend / Notifications / SLA

Description:

- FRT breach: email to assigned agent (or unassigned support queue) + in-app notification
- TTR breach: email to assigned agent AND supervisor (escalation_paths first step role) + SMS via Unifonic
- Email uses sla_frt_breach and sla_ttr_breach Resend templates
- In-app notification: stored in notifications table, shown in bell icon badge on dashboard
- Notification includes: ticket ID, subject, breach type, time overdue, direct link to ticket

Acceptance Criteria:

1. Breach alert sent only once per breach per ticket
2. Resend delivery confirmed (webhook or delivery status check)
3. In-app notification visible within 60 seconds of breach detection

9. Jira Tickets — Sprint 3: Notifications

PHYSIO-SUP-15 **Resend email templates: all 10 support notification templates (bilingual AR/EN)** SP: 5 | Sprint 3

Assignee: Full-Stack Dev | **Priority: High** | Tags: Support / Notifications / Resend

Description:

- Create 10 Resend React email templates: ticket_created_learner, ticket_assigned_agent, ticket_reply_notification, sla_frt_breach, sla_ttr_breach, ticket_resolved, csat_survey, nps_survey, csat_followup, ticket_escalation
- Each template supports Arabic (RTL) and English (LTR) based on user.language preference
- CSAT survey template: embeds 5 clickable star links (token-authenticated, one-click rating without login)
- NPS survey template: embeds 0–10 scale as clickable links
- All templates branded with Gamma Developers / PhysioGuidance visual style
- Variable interpolation: {{learner_name}}, {{course_name}}, {{ticket_id}}, {{frt_target}}, {{ttr_target}}, {{agent_name}}, {{survey_token}}

Acceptance Criteria:

1. All 10 templates render correctly in Gmail, Outlook, Apple Mail
2. RTL layout renders correctly for Arabic recipients
3. CSAT star links generate correct survey token and redirect to submission endpoint
4. Templates pass Resend send test without errors

PHYSIO-SUP-16 **Unifonic/Infobip adapter: SMS + WhatsApp notification service** SP: 5 | Sprint 3

Assignee: Backend Lead | **Priority: High** | Tags: Support / Backend / Notifications / Unifonic

Description:

- Implement NotificationService abstraction with channels: email (Resend), sms (Unifonic), whatsapp (Unifonic/Infobip)
- SMS messages: 160-character limit, Arabic and English templates
- WhatsApp messages: approved template messages (Unifonic template IDs stored in config)
- Channel selection based on: user.notification_preference (email/sms/whatsapp) + event type
- Mock adapter for development (Unifonic credentials pending from Amna)
- Retry logic: 3 attempts with exponential backoff for failed deliveries
- Delivery status logged to notification_logs table for audit

Acceptance Criteria:

1. Mock adapter returns success and logs correctly (unblocks development)
2. Channel fallback: if WhatsApp fails, fall back to SMS
3. SMS messages do not exceed 160 characters (Arabic counts double per char for encoding)
4. All delivery attempts logged with timestamp and status

PHYSIO-SUP-17	CSAT survey send job: async dispatch 15 minutes after ticket resolution	SP: 5 | Sprint 3
Assignee: Backend Lead	Priority: High	Tags: Support / Backend / CSAT / Notifications

Description:

- Background job queued when ticket status changes to resolved
- Job executes after 15-minute delay
- Survey token generated: signed JWT containing ticket_id, user_id, expires_in=72h
- Send via user.notification_preference channel (email default, WhatsApp if preferred)
- One CSAT survey per ticket (idempotent: check csat_surveys table before sending)
- POST /api/v1/surveys/csat endpoint: accepts token + score (1–5) + optional comment
- Score <= 2: queue csat_followup email

Acceptance Criteria:

1. Survey not sent if ticket was closed without resolution (e.g., duplicate/spam)
2. Expired token returns 410 Gone with message explaining survey expired
3. Score submission stores channel=email/sms/whatsapp correctly based on how user responded
4. Detractor follow-up (score <= 2) sent within 1 hour of submission

PHYSIO-SUP-18	NPS survey system: configurable send cadence + submission endpoint	SP: 5 | Sprint 3
Assignee: Full-Stack Dev	Priority: Medium	Tags: Support / Backend / Frontend / NPS

Description:

- Admin configuration: NPS send frequency (e.g., every 90 days per user per course, or post-completion)
- Batch NPS job: identify eligible users (completed course AND not received NPS in X days), dispatch survey
- NPS token: signed JWT with user_id, course_id, expires_in=7 days
- POST /api/v1/surveys/nps: accepts token + score (0–10) + optional comment
- Promoter (9–10), Passive (7–8), Detractor (0–6) classification stored
- GET /api/v1/surveys/summary: NPS score = (% Promoters - % Detractors), segmented by course/channel/period

Acceptance Criteria:

1. User not sent NPS more than once per configured interval
2. NPS net score calculated correctly: Promoters% - Detractors% (not including Passives in denominator)
3. Admin can preview NPS summary before sending batch

10. Jira Tickets — Sprint 4: CSAT/NPS Dashboard + Macros

PHYSIO-SUP-19 **CSAT & NPS results dashboard section on Support Dashboard UI** SP: 6 | Sprint 4

Assignee: Frontend Lead	**Priority: High**	Tags: Support / Frontend / CSAT / NPS / UI

Description:

- CSAT section: average score card (out of 5), score distribution bar chart (1–5), by-course breakdown table
- NPS section: Net Promoter Score card, Promoter/Passive/Detractor percentage donut chart, trend line by month
- Per-channel breakdown table: email vs SMS vs WhatsApp response rates and scores
- Verbatim comments feed: filterable by score range, course, date
- Date range filter + course filter
- Export CSAT/NPS data as CSV

Acceptance Criteria:

1. NPS score card shows correct formula: (Promoters% - Detractors%)
2. Verbatim comments paginated (not all loaded at once)
3. Charts render correctly with 0 responses (empty state shown)
4. Bilingual labels (AR/EN)

PHYSIO-SUP-20 **Macros CRUD backend + Apply Macro to ticket endpoint** SP: 5 | Sprint 4

Assignee: Backend Lead	**Priority: Medium**	Tags: Support / Backend / Macros / API

Description:

- Full CRUD for support_macros: GET list, POST create, PATCH update, DELETE
- POST /api/v1/tickets/:id/apply-macro: accepts macro_id, returns body with variables resolved
- Variable resolution: {{learner_name}} from ticket.user, {{course_name}} from ticket.course, {{ticket_id}} from ticket
- Macro can include kb_article_urls (array) — these appended as links in resolved body
- Support and Admin roles only can manage macros

Acceptance Criteria:

1. Unresolvable variables replaced with empty string (not literal {{var}} in output)
2. Apply-macro does NOT post the comment — only returns resolved text for agent to review
3. Deleted macro cannot be applied (404 on apply attempt)

PHYSIO-SUP-21	Macros & Templates UI on Support Dashboard	SP: 5 | Sprint 4
Assignee: Frontend Lead	Priority: Medium	Tags: Support / Frontend / Macros / UI

Description:

- Macros library page: list of macros with title, preview, category tag
- Create/Edit macro form: title, body (rich-text with variable hints), KB article URL array, escalation path link
- Apply macro from ticket detail view: dropdown selector showing macro titles, preview resolved text before inserting to reply composer
- Escalation paths configuration UI (linked to Sprint 2 backend)

Acceptance Criteria:

1. Variable hints shown in editor ({{}}-wrapped tokens with autocomplete)
2. Preview shows resolved output correctly
3. Bilingual: macro body can have AR and EN versions

PHYSIO-SUP-22	Knowledge Base CRUD (bilingual) + public-facing KB article pages	SP: 8 | Sprint 4
Assignee: Full-Stack Dev	Priority: Medium	Tags: Support / Backend / Frontend / KB

Description:

- Admin/Support: create and manage KB articles with Arabic and English content
- Categories: Billing, Login, Certificate, Content, Technical
- Public GET endpoint: /api/v1/kb/articles (no auth required)
- Public-facing KB page: searchable by keyword, filterable by category, language toggle AR/EN
- KB articles linked from CSAT survey email footer as self-service option
- Learner Dashboard FAQs section: shows top 5 articles by view_count per category
- Track view_count per article (increment on GET)

Acceptance Criteria:

1. Public page accessible without login (NELC requirement: user guides within platform)
2. Arabic content renders in RTL
3. Search works in both Arabic and English
4. WCAG 2.1 AA: accessible headings, search input labelled

11. Jira Tickets — Sprint 5: Reports + Cross-Dashboard Integrations

PHYSIO-SUP-23	Reports API: volume, top issues, SLA compliance, agent performance, export	SP: 6 | Sprint 5

Assignee: Backend Lead | **Priority: High** | Tags: Support / Backend / Reports / API

Description:

- GET /reports/support/volume: ticket count by day/week, broken down by category, priority, status
- GET /reports/support/issues: group tickets by category+subject keywords, return top 10 recurring with frequency
- GET /reports/support/sla-compliance: % tickets resolved within FRT and TTR SLA per period
- GET /reports/support/agent-performance: per-agent avg FRT, avg TTR, CSAT avg, tickets resolved
- GET /reports/support/export: generates CSV or PDF of selected report with timestamp, applied filters, generated_by user
- All reports support date_from, date_to, course_id, priority, category filters

Acceptance Criteria:

1. Top issues correctly aggregates similar subjects (basic keyword grouping)
2. Export PDF uses ReportLab with Gamma branding and NELC evidence header
3. Export includes: report_generated_at, filters_applied, generated_by fields for audit trail
4. All report endpoints restricted to Support/Admin roles

PHYSIO-SUP-24	Reports UI on Support Dashboard: volume chart, issues heatmap, SLA, agent table	SP: 6 | Sprint 5

Assignee: Frontend Lead | **Priority: High** | Tags: Support / Frontend / Reports / UI

Description:

- Reports section with sub-tabs: Volume / Top Issues / SLA Compliance / Agent Performance
- Volume tab: line/bar chart (daily/weekly toggle), breakdown by category as stacked bars
- Top Issues tab: horizontal bar chart or table sorted by frequency, category colour-coded
- SLA Compliance tab: gauge charts for FRT and TTR compliance %, breach trend line
- Agent Performance tab: sortable table with FRT avg, TTR avg, CSAT avg, tickets count
- Global filters: date range picker, course selector, priority filter
- Export button: triggers CSV or PDF download

Acceptance Criteria:

1. Charts render with Arabic-friendly fonts when language=AR
2. Export button disabled while export is generating, shows loading spinner
3. Empty states shown correctly (no data message) for new date ranges
4. All charts have accessible colour choices with pattern fills as WCAG backup

PHYSIO-SUP-25	Admin Dashboard: Support KPIs widget (FRT/TTR/CSAT/NPS/breach rate + issues heatmap)	SP: 6 | Sprint 5
Assignee: Full-Stack Dev	Priority: High	Tags: Support / Admin Dashboard / Frontend / Backend

Description:

- Add Support & Satisfaction widget section to Admin Dashboard
- KPI cards: Avg FRT (vs target), Avg TTR (vs target), Active Breaches, CSAT avg score, NPS score
- Common Issues heatmap: 2D grid of category (rows) x day-of-week (columns), colour intensity = volume
- Drill-down: click any KPI card to navigate to Support Dashboard with pre-applied filters
- GET /api/v1/admin/support-kpis endpoint aggregates all data in single call
- Data refreshed every 5 minutes

Acceptance Criteria:

1. Heatmap renders correctly with 0 values (grey/white cells, not broken)
2. Drill-down navigation passes date_from/date_to and category as query params to Support Dashboard
3. Admin role required (403 for other roles)
4. Widget shows 'No data' gracefully if no tickets in selected period

PHYSIO-SUP-26	Instructor Dashboard: 'Open Support Ticket' from At-Risk Learner intervention queue	SP: 5 | Sprint 5
Assignee: Full-Stack Dev	Priority: Medium	Tags: Support / Instructor Dashboard / Frontend / Backend

Description:

- Add 'Open Ticket' action button to each learner row in At-Risk Learners widget
- Pre-populates ticket form: subject = 'Learner Support: {learner_name} – {course_name}', category=content, priority=medium
- Pre-populates: user_id=learner_id, course_id, instructor context in ticket description
- Instructor receives copy of ticket confirmation (ticket_assigned_agent email with note that they opened it)
- Instructor can track status of tickets they opened via a 'Tickets I Opened' list

Acceptance Criteria:

1. Instructor cannot open ticket on behalf of a learner not in their course
2. Ticket creator stored as instructor_id in a ticket_opened_by field (separate from user_id which is the learner)
3. Learner notified that a support ticket was opened for them (ticket_created_learner email with note from instructor)

PHYSIO-SUP-27	Finance Dashboard: auto-generate support ticket on refund request / failed payment	SP: 5 | Sprint 5
Assignee: Full-Stack Dev	Priority: Medium	Tags: Support / Finance Dashboard / Backend

Description:

- When learner submits a refund request: auto-create ticket with category=billing, priority=high, invoice_id linked
- When payment fails (webhook from Tap/Moyasar): auto-create ticket if the failure persists after 2 retries, category=billing, priority=high
- Finance Manager sees open billing ticket count in Finance Dashboard header
- Billing tickets on Support Dashboard show invoice summary card in sidebar (amount, gateway, failure reason)
- GET /api/v1/tickets/course/:courseId/count: returns open ticket count badge for Course Manager

Acceptance Criteria:

1. Auto-created tickets have is_auto_generated=true flag for reporting
2. Duplicate prevention: don't create billing ticket if one already open for same invoice_id
3. Course Manager ticket count badge only shows tickets with matching course_id

PHYSIO-SUP-28	QA: End-to-end test suite for Support Dashboard (all features, all roles)	SP: 8 | Sprint 5
Assignee: QA Engineer	Priority: High	Tags: Support / QA

Description:

- Test all ticket lifecycle flows: create → assign → reply → resolve → close (all roles)
- SLA breach simulation: set FRT=1 min in test env, create ticket, verify breach fires within 2 cron cycles
- CSAT survey: resolve ticket, wait 15 min (or advance clock), verify survey email sent, submit score, verify stored
- NPS batch: trigger NPS job, verify eligible users receive survey
- Macro apply: create macro with variables, apply to ticket, verify variable resolution
- Reports: generate all 4 report types, verify CSV and PDF exports contain correct data and audit fields
- Cross-dashboard: verify Learner sees own tickets, Admin sees KPIs widget, Instructor opens ticket for learner
- RBAC: verify Learner cannot access Support Dashboard main board, cannot see internal notes
- WCAG 2.1 AA: automated axe-core scan on Tickets Board, Ticket Detail, Reports pages
- Bilingual: verify all UI renders correctly in Arabic RTL

Acceptance Criteria:

1. Zero accessibility violations at WCAG AA level on all pages (axe-core)
2. All 10 Resend templates confirmed delivered in test Resend account
3. SLA breach test fires within 10 minutes of threshold breach
4. All report exports contain: report_generated_at, filters_applied, generated_by
5. No 500 errors in any documented flow across all roles

12. NELC Compliance Mapping

NELC Requirement	Support Dashboard Coverage	Status
WCAG 2.1 AA Accessibility	All dashboard pages: colour contrast, keyboard nav, focus rings, ARIA labels, RTL layout. Tested via axe-cor	✓ Covered
Bilingual (Arabic/English) with RTL support	All UI components and email templates support AR (RTL) and EN (LTR) toggle. KB articles have bilingual col	✓ Covered
Learner support access (user guides within platform)	KB article system with public-facing pages. Learner Dashboard FAQs section. Accessible without login (PHYS	✓ Covered
Communication & Netiquette policy published	Link to policy in ticket creation form footer and KB category.	✓ Covered
Evidence export for audits (CSV/PDF with timestamps)	All reports exportable. Include generated_at, filters_applied, generated_by for NELC audit bundles (PHYSIO-SUP-	✓ Covered
Immutable audit logs	sla_events table is append-only. ticket status change history logged. notification_logs for all dispatched notific	✓ Covered
PDP Data Protection compliance	CSAT/NPS survey tokens expire in 72h/7d. Notification logs purge per retention policy. No PII in SLA breach	✓ Covered
e-Attendance and participation policy accessible	KB article category includes platform policies. Policy link in Learner Dashboard.	✓ Covered

13. QA Verification Checklist

Ticket Management

- [ ] Ticket creation works for all roles (Learner, Instructor, Support, Admin)
- [ ] Status transitions (New → In Progress → Resolved → Closed) work via API and drag-and-drop
- [ ] Internal notes not visible to Learner or Instructor roles
- [ ] first_response_at set on first public Support reply only
- [ ] CSAT job queued 15 min after resolution
- [ ] Learner cannot view another learner's ticket (403)
- [ ] Ticket from Instructor pre-populated with learner + course context

SLA Engine

- [ ] FRT breach fires correctly after frt_minutes with no response
- [ ] TTR breach fires correctly after ttr_minutes without resolution
- [ ] Breach notification sent only once per ticket per breach type
- [ ] Escalation path reassigns ticket after configured delay
- [ ] /sla/kpis averages are mathematically correct against test data
- [ ] Backlog buckets sum equals total open ticket count

Notifications

- [ ] All 10 Resend templates deliver successfully
- [ ] Arabic templates render in RTL in email clients
- [ ] CSAT one-click star links resolve score correctly
- [ ] NPS 0–10 links resolve score correctly
- [ ] CSAT survey not sent if ticket was closed as duplicate/spam
- [ ] Detractor follow-up (score <= 2) sent within 1 hour
- [ ] Unifonic mock adapter logs correctly (no live send in dev)

CSAT/NPS

- [ ] NPS net score = Promoters% - Detractors% (verified with manual calculation)
- [ ] User not sent NPS twice within configured interval
- [ ] Summary endpoint segments correctly by course and channel
- [ ] Expired survey token returns 410 with message

Macros & KB

- [ ] Variable substitution resolves correctly ({{learner_name}}, {{course_name}}, {{ticket_id}})
- [ ] Unresolvable variables replaced with empty string
- [ ] Apply-macro returns resolved text only (does not post comment)
- [ ] KB articles publicly accessible without login
- [ ] Arabic KB articles render in RTL

Reports & Exports

- [ ] All 4 report types return correct data for test dataset
- [ ] CSV export contains correct headers and rows
- [ ] PDF export contains: generated_at, filters_applied, generated_by
- [ ] Export restricted to Support/Admin roles

Cross-Dashboard

- [ ] Admin KPIs widget loads correctly with correct aggregates

- [ ] Instructor open-ticket flow populates all context fields
- [ ] Learner ticket count and status visible on Learner Dashboard
- [ ] Finance billing ticket auto-created on refund request
- [ ] Duplicate billing ticket prevention works (same invoice_id)

Accessibility (WCAG 2.1 AA)

- [ ] Tickets Board: axe-core 0 violations
- [ ] Ticket Detail: axe-core 0 violations
- [ ] CSAT/NPS section: axe-core 0 violations
- [ ] Reports section: axe-core 0 violations
- [ ] All interactive elements keyboard-reachable (Tab order logical)
- [ ] Priority badges: colour contrast ratio >= 4.5:1
- [ ] Arabic RTL layout verified on all pages